Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/06/2009
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	djdurian@physics.upenn.edu	**Fax:**	FY 215-898-2010
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-8147
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	209 South 33rd Street		
PI Web Page:	http://www.physics.upenn.edu/		
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	04/04/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
Contact Email:	DeVon.Griffin@grc.nasa.gov , DeVon.W.Griffin@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this period my group included postdoc Chris Jones, Penn graduate student Adam Roth, UCLA graduate student Alex Gittings, and visiting summer Cornell graduate student Jennifer Reiser. During the past year we have made significant progress on several different aspects of the overall FOAM project: (1) First I helped formulate a new ESR (Experimental Requirements Document), which is now nearly ready to go out to engineering, where the focus is shifting in the near term to coarsening. This is not a new direction, but rather represents a subset of the originally envisioned and funded ultimate microgravity experiments. The restricted experiment features optical probes, like DTS, DWS, and now SVS as crucial diagnostics of the structure and dynamics. (2) In terms of communicating results, I gave invited presentations, both as a plenary speaker at the conference Dynamics Days 2008 in Delft, and as a technical seminar at the research lab of Rhodia corporation in Bristol PA. Furthermore, I published two papers and mentored a student speaker, and I acted as co-organizer of EuFoam 2008 in Noordwijk. Research accomplishments fall into several categories: (4) Speckle-Visibility Spectroscopy. In this thrust we finished our analysis, wrote a paper, and revised it in response to referee reports; it has now appeared. (5) Foam Optics. Here we are revisiting our prior simulations of light propagation inside a foam structure, now incorporating the effects of thin-film interference when light reflects foam the soap films. This work will complete the PhD thesis of Alex Gittings, and should be finished by summer 2009. (6) Rheology. Here we are attempting a novel rheological diagnostic based on measurement of velocity profile in a channel subject to a known pressure gradient. This method, if successful, could prove useful in microgravity where it is proving difficult for engineering and fiscal reasons to implement a more conventional rheometer. Preliminary results are very promising, to the point that we are giving a contributed talk at the APS March Meeting this year. (7) Tomography. We have succeeded in taking extremely high quality photos of a foam sample from many angles, and we have succeeded in reconstructing a three-dimensional map of structure. What remains is to interpret this by identifying the locations of topological features of vertices and Plateau borders. This is proving difficult in 3d. Therefore we are attempting the same in 2d. (8) Coarsening. As part of 2d/3d foam imaging project we are carefully following the time evolution as gas diffuses from small to large bubbles. So far we are having good preliminary success in being able to test von Neumann's law, that the rate of bubble area change in proportional to the number of sides minus six. Our tests are of higher precision than any in the literature. In addition, our unique sample cells allow us to vary the liquid fraction via external pressure. And furthermore we are investigating the influence of curvature, by embedding the foam between clear plastic hemispheres. (9) Finally I am undertaking a realistic calculation of the partition of liquid between vertices, borders, and films by

	detailed balance of Laplace and disjoining pressures for carefully known miscrostructure. This addresses the widely-made assumption that liquid resides primarily in borders and that film thickness is constant, which underly our baseline thinking and our analysis of eg electrical conductivity data in terms of liquid fraction.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Jones CD, Nordstrom K, Durian DJ. "Pressure-driven foam flow rheology." Presented at the Annual March Meeting of the American Physical Society, Pittsburgh, PA, March 16-20, 2009. Bulletin of the American Physical Society 2009 Mar;54(1):P14.00008. http://meetings.aps.org/Meeting/MAR09/Event/96915 , Mar-2009
Articles in Peer-reviewed Journals	Feitosa K, Durian DJ. "Gas and liquid transport in steady-state aqueous foam." Eur Phys J E Soft Matter. 2008 Jul;26(3):309-16. PMID: 18516492 ; http://dx.doi.org/10.1140/epje/i2007-10329-6 , Jul-2008
Articles in Peer-reviewed Journals	Gittings AS, Durian DJ. "Statistics of bubble rearrangement dynamics in a coarsening foam." Phys Rev E Stat Nonlin Soft Matter Phys. 2008 Dec;78(6 Pt 2):066313. PMID: 19256951 ; http://dx.doi.org/10.1103/PhysRevE.78.066313 , Dec-2008